

Project Proposal: Budget-Aware DPO

Democratizing Adaptive Reasoning via Token-Efficient Distillation

Research Question

Can a 0.5B parameter model maintain high mathematical accuracy while significantly optimizing computational latency through **complexity-aware distillation**? Specifically, can **Direct Preference Optimization (DPO)** be used to train a student model to adapt its reasoning length—prioritizing concise "System 1" responses for simple arithmetic while retaining detailed "System 2" Chains-of-Thought (CoT) for complex mathematical proofs?

Motivation & Related Work

Standard distillation techniques typically force small models to mimic the highly verbose reasoning paths produced by massive teacher models (e.g., Llama-3.1-405B). While these long chains-of-thought improve accuracy, they introduce significant token redundancy, "over-reasoning" on trivial tasks, and high inference costs that are prohibitive for edge deployment.

Our project explores **Reasoning Density Optimization**. We aim to prove that a small-scale model can internalize logically sound but compressed paths for simple problems while reserving its "computational budget" for complex ones. This mimics a human-like routing between intuitive and analytical thought, avoiding the repetitive loops common in standard distilled models.

Methodology & Requirements

1. Data Preprocessing: The 4-Way Augmentation Matrix

We will utilize the **OpenMathInstruct-2** dataset (14M teacher-generated solutions). To enable adaptive reasoning, we will implement a multi-stage preprocessing pipeline to classify problems into a complexity matrix:

- **Classification Criteria:** Problems are labeled as **Easy** (e.g., GSM8K, low teacher token count) or **Hard** (e.g., MATH Level 5, high teacher token count).
- **Augmentation Strategy:**
 - **Easy-Correct:** Short, direct-to-answer sequences are labeled as **Preferred**. Verbose or redundant correct paths are labeled as **Rejected**.
 - **Hard-Correct:** Detailed, logically dense CoT paths are labeled as **Preferred**. Oversimplified or brief answers (even if correct) are labeled as **Rejected**.
 - **Incorrect Answers:** Any logically flawed path is labeled as **Rejected** across all

complexity levels.

2. The Budget-Aware DPO Loss Function

Instead of a uniform preference for correctness, our DPO objective is conditioned on the **Complexity Flag (\$C\$)**. The standard DPO loss is defined as:

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim D} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w|x)}{\pi_{\text{ref}}(y_w|x)} - \beta \log \frac{\pi_{\theta}(y_l|x)}{\pi_{\text{ref}}(y_l|x)} \right) \right]$$

To achieve token efficiency, we modify the implicit reward $R(x, y)$ by introducing a **Length-Penalty Term (\$\lambda\$)** that scales with complexity:

$$R_{\text{budget}}(x, y) = \beta \log \frac{\pi_{\theta}(y|x)}{\pi_{\text{ref}}(y|x)} - \lambda(C) \cdot |y|$$

- **For Easy Tasks (\$C=0\$):** λ is high. The model is heavily penalized for every unnecessary token, forcing it to shift probability mass toward concise, direct logic.
- **For Hard Tasks (\$C=1\$):** λ is near zero. Accuracy is the primary objective, allowing the model to utilize full CoT to solve complex calculus or differential equations.

3. Technical Stack

- **Student Model:** Qwen-2.5-0.5B (chosen for high baseline reasoning density).
- **Hardware:** NVIDIA RTX 6000 Ada (48GB VRAM).
- **Tools:** Unsloth or Axolotl for memory-efficient training; vLLM for inference benchmarking. We will perform full-parameter training or high-rank LoRA ($r=128+$) to ensure stable convergence.

Evaluation & Expected Outcomes

We will measure the **Pareto frontier** between mathematical accuracy and computational efficiency compared to standard SFT and DPO baselines.

1. Accuracy & Efficiency Metrics

- **Accuracy:** Evaluated on **GSM8K** (Easy) and **MATH** (Hard). We expect to maintain **>95%** of the baseline accuracy on MATH Level 4-5 while reducing total tokens.
- **Tokens Per Correct Answer (TPCA):** We hypothesize a **30-50% reduction** in TPCA on GSM8K by eliminating "over-reasoning."
- **Inference Latency:** Real-world time-per-query reduction measured via vLLM.

2. Qualitative Distribution Analysis

We expect to observe a **bimodal distribution** in response lengths. The model should demonstrate "Learned Routing"—naturally producing 20-40 tokens for simple arithmetic and

300+ tokens for complex proofs—confirming it has internalized an "intuitive" vs. "analytical" reasoning style without a hard-coded router.

Metric	Baseline (Distilled)	Budget-Aware DPO
Avg. Tokens (Easy)	High (~150)	Low (<40)
Avg. Tokens (Hard)	High (~300)	High (~300)
Reasoning Style	Verbose / Repetitive	Adaptive / Concise

Bibliography

- NVIDIA. (2024). *OpenMathInstruct-2: Accelerating AI for Math with Large-scale Synthetic Data*.
- Alibaba Cloud. (2024). *Qwen2.5-Math: Toward Mathematical Expert at 7B and Below*.
- Rafailov, R., et al. (2023). *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*.

AI Usage Disclosure: Generative AI was used for formatting and structural refinement. Experimental design, complexity-aware routing logic, and technical methodology were developed by the authors.

Would you like me to generate a **Python script** for the preprocessing logic to automate the 4-way classification based on your specific token thresholds?